

Google Trends in Depth

From Raw Search Data to Disease Forecasting and Population Insight

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Digital search as a population health signal

- Traditional surveillance (lab confirmation, syndromic reporting) is rigorous but slow: testing, reporting, and aggregation introduce delays of days to weeks, precisely when early warning matters most.
- People often turn to the internet for health information before, or alongside, consulting a healthcare provider, making search activity an early signal of health-seeking behavior.
- Billions of daily online searches leave a digital footprint of public interests and behaviors, captured continuously and at population scale.

Google Trends has become a popular data source for infectious disease modeling, but using it effectively requires answering three fundamental questions.

1. What is Google Trends data, and how is it generated?
2. How can we transform noisy, distorted search data into reliable predictive signals?
3. What can search behavior reveal about disease dynamics beyond prediction?

What is Google Trends?

Understanding Google Trends

Statistical Modeling of Google Trends

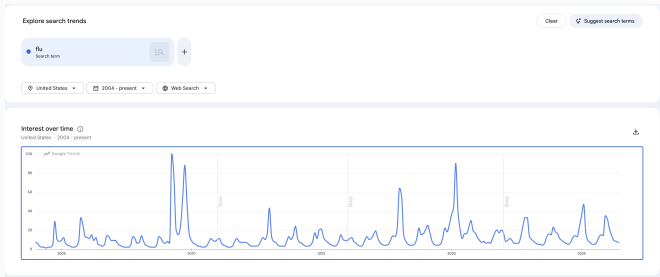
Forecasting with Google Trends

Beyond individual time series: temporal structure of search behavior

Conclusion

What is Google Trends?

The interface: interest over time

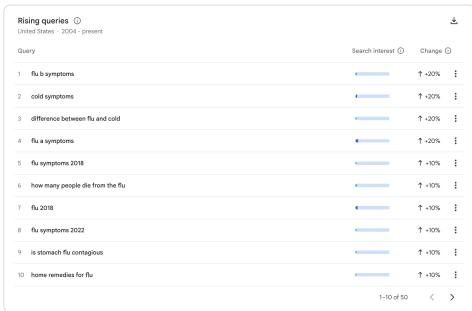
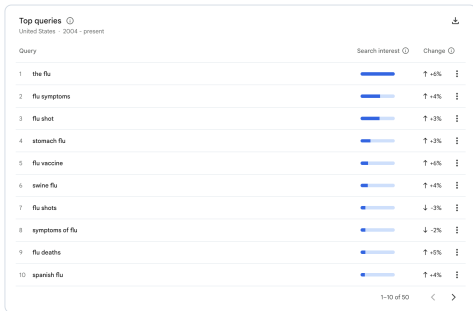


- Tool that provides insights into the popularity of search terms across different periods and regions.
- Users can analyze search behavior by location, time range, category, type of search.
- *Interest over time*: index of relative search volumes of a query, revealing trends, seasonal patterns, and spikes.

The interface: related queries

Commonly searched queries

People who searched for **flu** also searched for these queries



- *Commonly searched queries*: terms searched alongside the query of interest, split into **top queries** (most popular terms in the period, location, category chosen) and **rising queries** (emerging terms that had a significant increase in search volume).
- Basis for building a term pool (next section).

- **Category:** restricting to “Health” removes unrelated noise.
- **Geography:** state- and metro-level pulls are noisier and reach the zero threshold more often than national pulls.
- **Time window:** a shorter window re-normalizes 0–100 over fewer points, changing the reported values even for identical underlying searches.
- **Search type:** web, image, news, and YouTube search. Most common is web search.
- Each of these settings redefines the normalization; pulls should be kept consistent when compared.

Why Google Trends?

Google Trends is a valuable tool:

- Free, real-time, publicly available time series of Google search activity since 2004. Widely used in research and industry as a leading indicator that complements traditional observed data.
- In the context of public health, individuals often turn to the internet first for information on symptoms and treatments, ahead of formal medical contact.
- Symptom or treatment-related queries (e.g., “cough”, “fever”, “flu symptoms”) often precede increases in clinical activity.
- The first use of Google search data was to forecast influenza outbreaks with Google Flu Trends (Nature paper, 2009).

Discussion

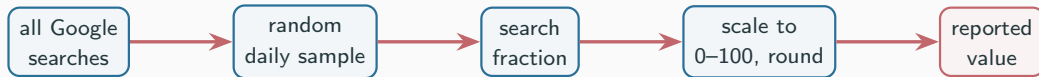
Have you used Google Trends before?

Understanding Google Trends

Data generation mechanism

Google Trends does not report the number of searches. It reports **relative interest**.

- Values are scaled **0 to 100** *within* the query set, time window, and geography requested. Changing any of those changes every value in the series.
- It is computed from a **random sample** of Google searches, re-drawn every day. The same query pulled on two different days can return different numbers.
- Low search volumes are rounded, and below an undisclosed privacy threshold they are reported as **zero**.



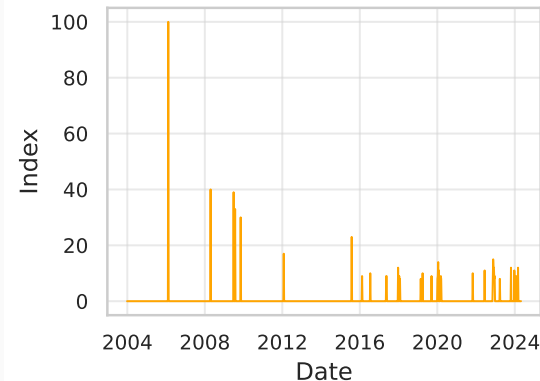
A single raw pull is a noisy, rescaled, partly censored realization, not a ground-truth measurement.

The data generation mechanism introduces four distortions

1. **Missing values** (zeros from the privacy threshold),
2. **Sampling variability** (different values across downloads),
3. **Noise** (within-sample jitter),
4. **Trend** (magnitudes shift).

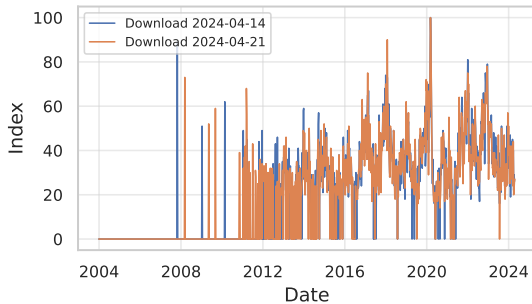
These distortions propagate directly into forecasting models and are rarely explicitly corrected in practice.

Distortion 1: missing values



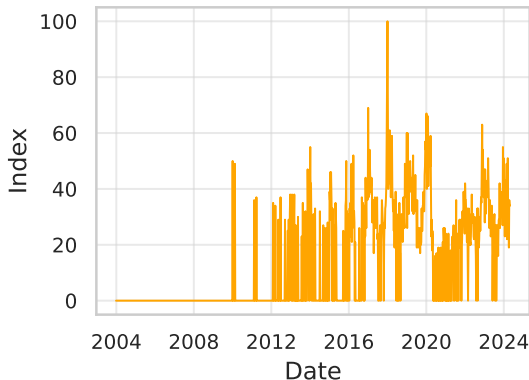
- An undisclosed privacy threshold censors search volumes to zero when searches are too sparse in a region to maintain user anonymity.
- A value of 0 indicates low search volume, not the absence of searches.
- Indicator: a column with more than 80% zeros.

Distortion 2: sampling variability



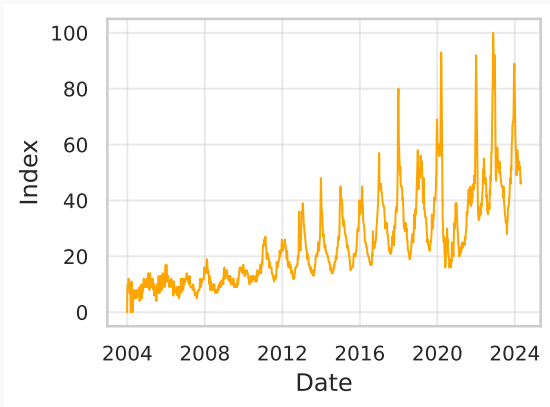
- Google servers cache results for 24 hours; at midnight UTC the cache expires and a new random sample is drawn.
- Repeated downloads of the same query, on different days, produce different realizations of the series, including for historical dates.
- This produces across-download variability that makes it difficult for reproducibility.

Distortion 3: noise



- Billions of daily Google searches occur; Google Trends provides only a representative sample, which is a random approximation of the full population of searches.
- This approximation contains within-sample noise unrelated to variation in search behavior.
- Low signal-to-noise ratio can obscure the underlying signal.

Distortion 4a: trend

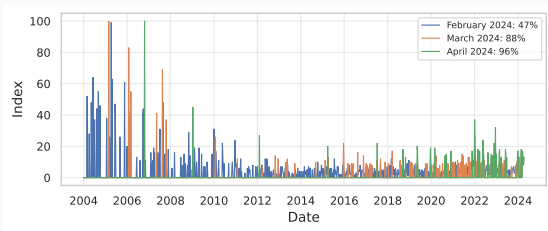


- Popularity of a topic changes with rising or declining public interest; vocabulary itself changes (new terms, obsolete ones).
- Platform's dynamic normalization that scales each series to 0–100 within the requested time window can introduce artificial structural breaks.
- Result: a long-term upward or downward trend that does not reflect actual changes in search behavior.

Distortion 4b: algorithm updates



Google annotates the site itself when the data collection pipeline is updated.



Data quality has also deteriorated in recent years: more zeros and more noise, even for previously stable queries. 15/39

Statistical Modeling of Google Trends

1. Methodology:

- Develop a reproducible 3-step preprocessing pipeline
- That restores the quality of Google Trends data for real-time forecasting

2. Empirical evaluation:

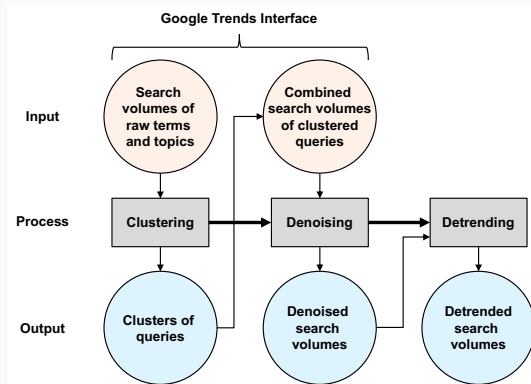
- Demonstrate improvements in influenza hospitalization forecasting
- Across statistical and machine learning models



We applied our pipeline to real-world, real-time data in the CDC FluSight Challenge to forecast weekly influenza hospital admissions in the 2024–2026 seasons.

Pipeline Overview: three corrections, matched to the four distortions

Methodology: **hierarchical clustering, smoothing splines denoising, detrending.**



Each step addresses a specific distortion.

Step 1: hierarchical clustering

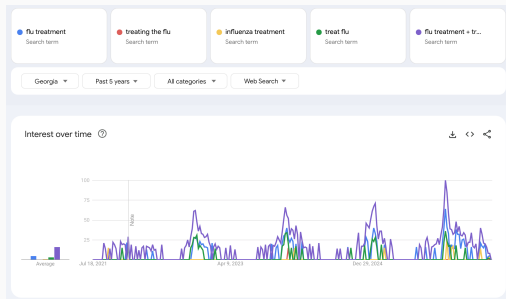
Goal: Reduce missing values by aggregating semantically related terms

- Cluster queries based on correlations between their time series
- Select the number of clusters via the elbow method
- Aggregate queries from each cluster using Google Trends “+” operator
- Retrieve combined search volumes

Evaluation: does aggregation reduce sparsity?

- 28,000+ state-level queries grouped into 300+ clusters across locations
- Metric: percentage of zeros before vs. after clustering

Combine terms with Boolean “+”



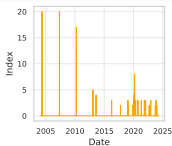
- "flu treatment" + "treating flu" is interpreted as "flu treatment" OR "treating flu" and returns one composite series in place of two separate, sparser ones.
- Search volumes that are mostly zero for an individual term can become a usable signal once combined.

Discussion

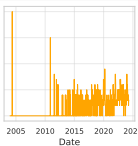
If you had 50 candidate search terms for your disease of interest, how would you decide which ones to group together? What criteria would you use?

Combining related terms effectively reduces sparsity

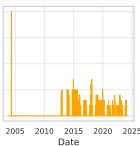
“cold virus”
97% zeros



“high fever”
64% zeros

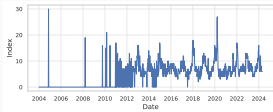


“symptoms of pneumonia”
87% zeros



“cold virus + high fever + symptoms of pneumonia”

Combined search volumes: 39% zeros



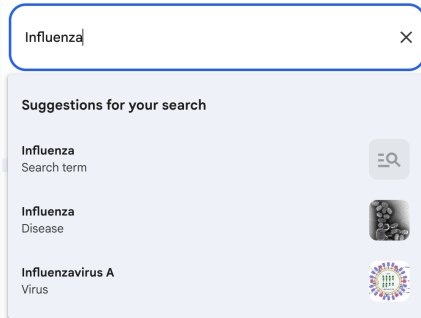
Summary across our entire dataset:

Before	Most search terms
clustering	have > 90% zeros
After	Most clusters fall
clustering	in the 0–30% zeros
	range

Clustering raises combined volumes above Google’s privacy thresholds.

Alternative: use topics, not raw strings

- A raw string (“flu”) misses alternate spellings, synonyms, and other languages.
- Google Trends **topics** aggregate semantically equivalent searches across spellings and languages into a single entity.
- Topics carry more volume and fewer zeros, which directly mitigates Distortion 1 (missing values). Topics should be preferred whenever they exist.



Step 2: smoothing splines denoising

Goal: Mitigate sampling noise and variability by estimating the underlying signal

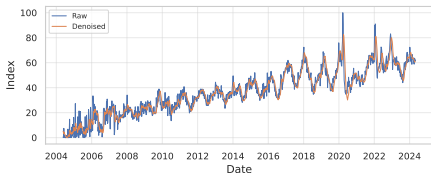
- Generate one-step-ahead smoothing spline predictions of search volumes
- Optimize smoothing parameters by minimizing the RMSE
- Fit splines on a 20-week rolling window using only recent past data
- Ensure a fully real-time implementation without look-ahead bias

Evaluation: does denoising improve signal quality?

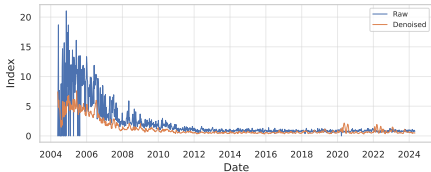
- ~30 downloads of each query retrieved
- Metric: Signal-to-Noise Ratio (SNR) = mean / SD across downloads
 - Measures how strong the average signal is compared to its variability
 - Compare SNR before and after denoising
 - Higher SNR indicates a stronger signal relative to sampling variability

Denoising reduces noise and sampling variability

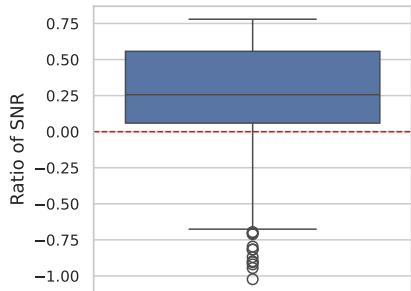
Mean = signal across 30 downloads for before and after denoising



Standard deviation = noise



Summary across our entire dataset:
Ratio of SNR = $\text{SNR}(\text{denoised}) / \text{SNR}(\text{raw})$



Denoising consistently improves signal quality.

Step 3: detrending

Goal: Remove long-term trends by eliminating non-stationary components

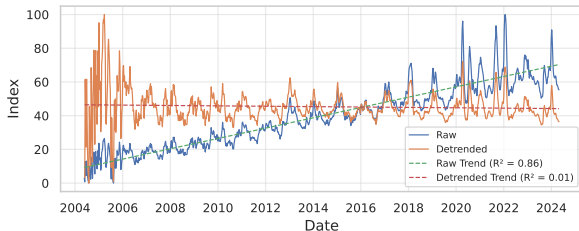
- Identify presence of trend using the Augmented Dickey-Fuller (ADF) test
- Estimate trend parameters using only the training set
- Remove deterministic trends via linear or quadratic detrending
- Difference stochastic trends to achieve stationarity

Evaluation: does detrending remove long-term trends?

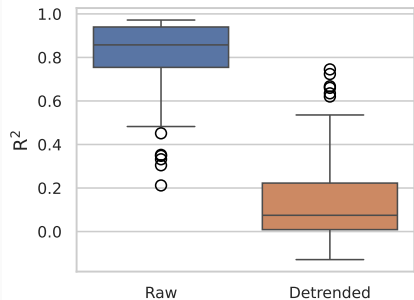
- Apply detrending to queries identified as non-stationary by the ADF test
- Metric: R^2 of fitted trend before vs. after detrending
 - Lower R^2 indicates successful trend removal

Detrending eliminates long-term trends

Estimated deterministic trend in raw and detrended data



Summary across our entire dataset:
mean trend R^2 drops from 0.82 to 0.14



Detrending restores stationarity and isolates the short-term signal.

Forecasting with Google Trends

Forecasting influenza hospitalizations using Google Trends

Goal: evaluate whether preprocessing improves the forecasting value of Google Trends as a predictive signal.

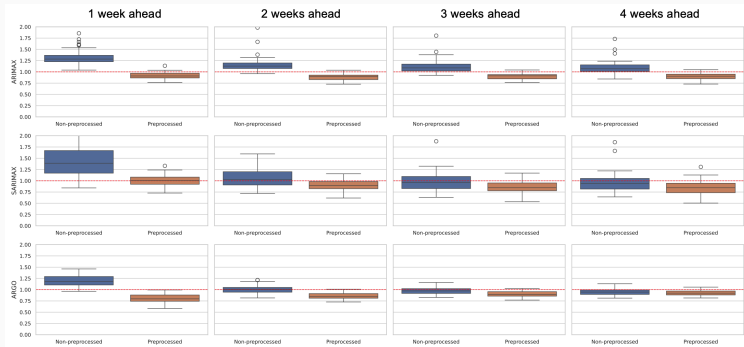
- **Task:** 1–4 week-ahead, state-level, out-of-sample forecasts over the 2022–2024 flu seasons.
- **Exogenous predictors:** influenza-related Google Trends search terms.
- **Target:** CDC influenza hospitalization counts.
- **Evaluation:** RMSE between forecasts and observed hospitalizations.
- **Comparison:** raw vs. preprocessed Google Trends vs. no search data (baseline).

5 forecasting models, selected to be interpretable enough to isolate Google Trends' contribution:

- Statistical: ARIMAX, Seasonal ARIMAX, ARGO.
- Machine learning: LightGBM, AdaBoost.

Forecasting results with statistical models

Relative RMSE across U.S. states (baseline = 1.0)



Non-preprocessed (raw) data above the baseline (in red) degrades the performance.

Preprocessed below the baseline improves performance across all 3 models and 4 horizons.

Preprocessing is necessary: statistical models rely heavily on exogenous inputs for predictive information.

Panel Diebold-Mariano test validates visual observations

* = $p < 0.05$ for models with exogenous data against models omitting search data

– = $p > 0.95$ for models worse than the baseline

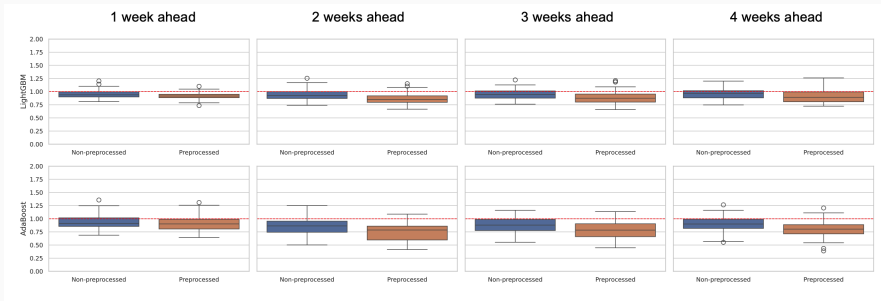
	1 wk		2 wks		3 wks		4 wks	
ARIMAX	–	*	–	*	–	*	–	*
SARIMAX	–	–				*	–	*
ARGO	–	*		*		*		*
LightGBM	*	*						
AdaBoost			*	*		*		*

■ Non-preprocessed ■ Preprocessed

Raw Google Trends data significantly degrades performance in statistical models compared to omitting search data.

Forecasting results with machine learning models

Relative RMSE across U.S. states (baseline = 1.0)



Non-preprocessed data below the baseline improves performance even without preprocessing.

Preprocessed below the baseline further improves performance across models and horizons.

Preprocessing is beneficial but not critical: ML models are less sensitive to distortions because of internal feature selection and regularization.

Wilcoxon signed-rank test validates visual observations

* = $p < 0.05$ for models with exogenous data against models omitting search data

	1 wk		2 wks		3 wks		4 wks	
ARIMAX		*		*		*		*
SARIMAX				*		*		*
ARGO		*		*	*	*	*	*
LightGBM	*	*	*	*	*	*	*	*
AdaBoost	*	*	*	*	*	*	*	*

■ Non-preprocessed ■ Preprocessed

Preprocessed data significantly improves performance across all models compared to omitting search data.

Discussion

In your experience with other exogenous variables, have you seen similar patterns where auxiliary data hurts statistical models but ML models still extract value?

- So far: statistical preprocessing restores the predictive power of individual search terms, improving forecasts of clinical activity.
- But each term (“flu”, “fever”) is treated in isolation, as a single noisy time series to be cleaned and fed into a model.
- Can the *structure* of search behavior itself, across a whole family of related terms, teach us something about disease dynamics?

Beyond individual time series: temporal structure of search behavior

- The first part restored the quality of individual terms' time series. However, a single query ("flu") only captures a slice of behavior. An outbreak is reflected across a family of queries: prevention, symptoms, treatment, complications.
- We evaluate searches from two complementary platforms:
 - **Google Trends** (population-level public search, triggered by personal experience of illness)
 - **UpToDate** (clinician decision support, triggered by patient encounters)
- Together they capture the full life cycle of a respiratory disease, from community spread to clinical management.

Beyond a single term's time series: the query family carries its own temporal structure.

Data and contributions

Weekly, state-level respiratory disease signals, 50 U.S. states, two seasons (2022–2024):

- **Google Trends:** 36 public topics.
- **UpToDate:** 21 clinical concepts.
- **Outcome:** Emergency Department visits for influenza and RSV (CDC NSSP).

First systematic test across two data sources and two pathogens:

1. A three-stage temporal structure: anticipatory searches lead, concurrent searches are contemporaneous, lagging searches follow.
2. Dual data-source comparison: public vs. clinician searches, same diseases.
3. Two-pathogen framework: influenza and RSV, identical methodology.

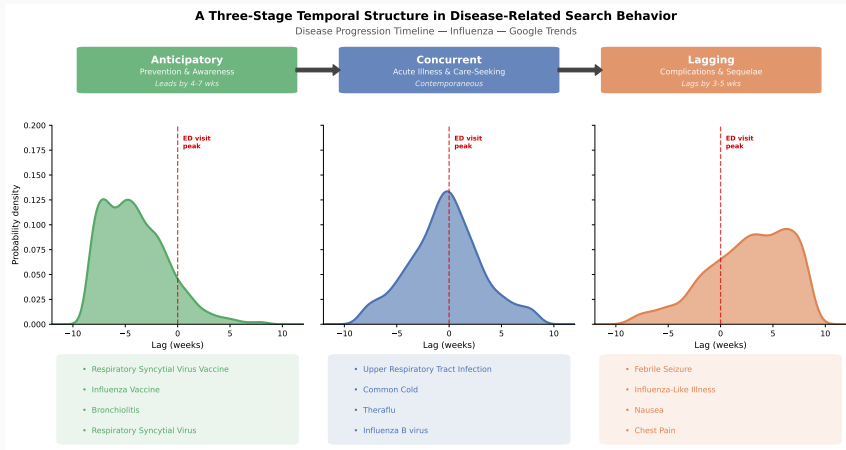
Keyword clustering based on optimal lags relative to ED visits

1. **Measure timing:** for each term in each state, compute the Pearson correlation with ED visits at each of 17 lag offsets (-8 to $+8$ weeks); convert to a probability distribution via softmax to emphasize the lags with strongest correlation.
2. **Aggregate across states:** average the 50 state-level distributions into one national timing profile per term, weighted by signal strength, retaining the full distribution rather than a single best lag. Result: one timing profile per search term.
3. **Group by timing:** Cluster terms by the similarity of their timing profiles using hierarchical clustering.

Discussion

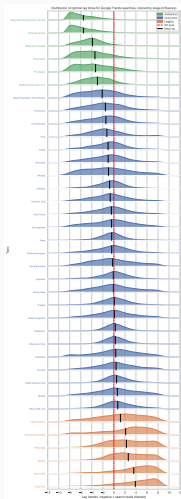
What clusters of searches would you expect around a respiratory illness? How many, and what terms would fall into each?

When do people search for respiratory-related topics relative to when they actually show up in the ED?

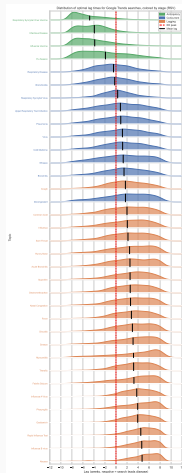


Probability that a group of search terms peaks at a given lag relative to the ED visit peak.

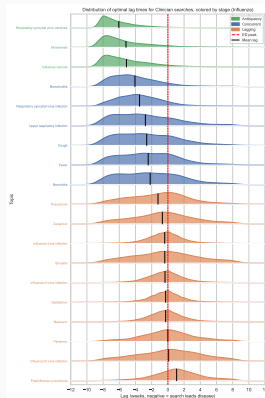
Generalizability: Consistent Structure Across Data Sources and Diseases



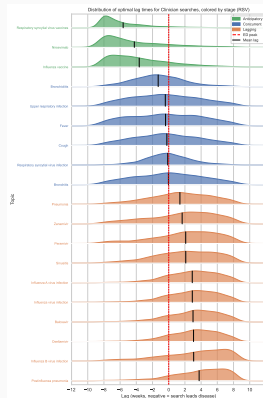
Google Trends –
Influenza



Google Trends –
RSV



Clinician – Influenza

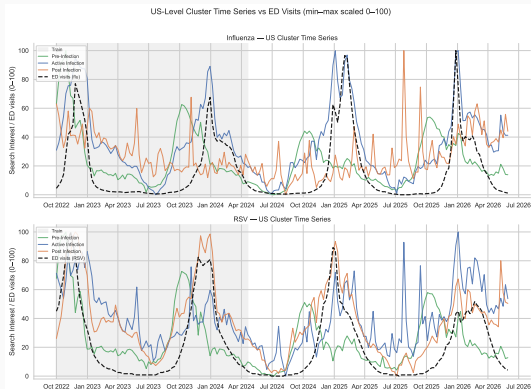


Clinician – RSV

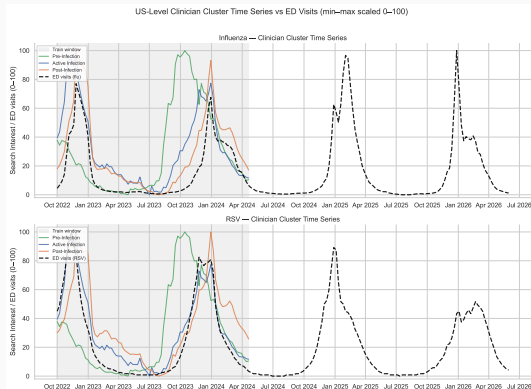
Discussion

Look at how the three clusters track the ED curve. What would you use each cluster for operationally?

Cluster composites track disease activity in two pathogens



Google Trends



Clinician

US-level Anticipatory/Concurrent/Lagging composites vs. ED visits, influenza and RSV, across four full seasons. The same structure that emerges in public search also emerges independently in clinician search.

Conclusion

Forecasting performance depends on the quality of input signals

Using raw Google Trends data:

- Degrades statistical models compared to omitting search data
- Improves machine learning models, but less than our approach

Preprocessing significantly improves accuracy across all 5 models and 4 forecast horizons.

Each pipeline step addresses a known distortion, moving beyond ad hoc feature engineering.

Findings generalize beyond Google Trends to any platform-generated behavioral signal.

Why the stage structure matters

- This is not simply “search predicts cases.” The category of a search term indicates which stage of the outbreak is underway.
- The 3-stage structure emerges across 4 disease-source combinations, indicating a real epidemiological structure.
- Implications for forecasting:
 - Leading indicators: anticipatory terms provide 4–7 weeks of advance signal.
 - Nowcasting: concurrent terms track current burden in real time.
 - Signal separation: stage-aware term selection can improve predictions.

Accounting for these stages may improve the interpretation of digital health signals, feature selection for forecasting models, timeliness of epidemic surveillance.

Questions?